Project 1 Launch: Proposal & Dataset Exploration

Problem Definition

Business goal

Our goal is this project will be to detect potential fraud in retail transactions. Fraud can look like unusually large purchases, suspicious pricing, or repeated orders that inflate revenue. By finding these patterns early, the business can reduce financial losses, protect customers, and keep trustworthy reports.

The pipeline of this project will be first to scan a large order dataset, second to find unusual transactions, and third to create a clear output that a team can review.

Fraud hurts a retail business and by reducing real revenue, they will have to pay back or refunds. Customers may stop buying if they feel unsafe. Importantly, fake transactions can distort real revenue and performance metrics. Detecting fraud is not only about catching criminals, it's also about keeping the business data reliable so decisions are based on truth.

Stakeholders

The security or risk team is the main user of this system. They can review flagged transactions and investigate suspicious activity. They can decide whether an order should be blocked, refunded or reported. They need clear anomaly results supported by evidence.

The finance and accounting team needs to have correct revenue reporting. Fraudulent transactions can distort financial reports and lead to incorrect business decisions. They will use fraud signals to reduce incorrect totals and track potential losses.

The operations team is also involved. If an order is suspicious, they may need to pause shipping, verify details, and reduce losses before the product leaves the warehouse.

Dataset Overview

Our dataset retail_orders_large.csv contains 75000 rows and 8 columns. The data covers transactions from January 1, 2022, to June 18, 2024, and it includes 4 regions (East, West, North, South). There are 899 unique products and 50912 unique customers.

Size and Data Quality

There are no missing values in any column, and there are 0 duplicate rows. The revenue column matches quantity * unit_price. The values also look reasonable, between 1 and 5, unit price is between $5.01 and $499.99, and revenue is between $5.01 and $2499.65. In real retail systems, data quality includes incorrect data types, missing prices, duplicated orders, negative or zero values, and strange outliers. That is why the notebook and our files in /src still include checks, conversions, and safe error handling.

Why Large Datasets Matter for Analytics

Our dataset is big enough to show performance problems if the code is inefficient. Fraud detection and analytics often require operations like grouping, clustering, sorting, rolling windows, and anomaly detection. Large datasets matter because they force us to use efficient methods and to design algorithms that scale. This is also important because if we want to use Machine Learning models, datasets size matter. The more data we have, the more models are able to detect patterns. This is important for fraud detection because classes might be unbalanced.

Algorithm Plan

In our project, we will use four algorithms because fraud doesn't appear in only one form. Some fraud looks like "big money" activity, some looks like repeated transactions, and some appears as sudden changes over time. Each algorithm gives a different point of view of risk, and together, they make the detection better.

1) Top-K Revenue Product

This algorithm dins the products that generate the most total revenue. It's useful for a fraud detection system because products that generate high revenue are often targeted. If a small set of products produces unusually large revenue, the security team can check those items. It also helps answer questions like "Which products should be checked more often?" and "Are we seeing abnormal revenue based on few products?"

2) Duplicate Detection

Duplicate detection finds rows that appear more than once. Duplicates can be a signal of fraud or data problems. For example, a transaction could be accidentally processed twice, or a fraud. Even if duplicated are not fraud, they still create false revenue totals, which can hide or confuse fraud patterns. This algorithm helps keep the dataset trustworthy before running other analysis.

3) Rolling Averages

Rolling averages compute daily revenue and then smooth it over a time window. This is needed because raw daily revenue can be noisy. A rolling average makes trends easier to see and helps detect unusual spikes or sudden changes. If revenue jumps for several days, that could be a signal of suspicious activity, a coordinated fraud event, or abnormal buying behavior that needs review.

4) Anormaly Detection

The z-score method flags what we call outliers. In our case, our outliers are transactions with revenue far from the normal range. This is one of the most direct fraud signals in our project. If an order's revenue is extremely high compared to the rest of the dataset, it may represent suspicious behavior such as stolen cards etc. Z-score anomaly detection gives a clear "why it was flagged" value, which is helpful for humans reviewing the results.

Security & Ethics

Logging

Logging keeps a record of what the program did, like loading the file and how many duplicates or anomalies were found. This helps with debugging and audits

Error Handling

Real data can have issues, such as missing files, wrong columns, bad dates). Try/except and input checks stop crashes and make errors clear and safe

Responsible AI Usage

An anomaly is only a “warning”, not a proof of fraud. Results should be reviewed by a human to enhance customer experience, and the data should be handled carefully to avoid sharing sensitive information.